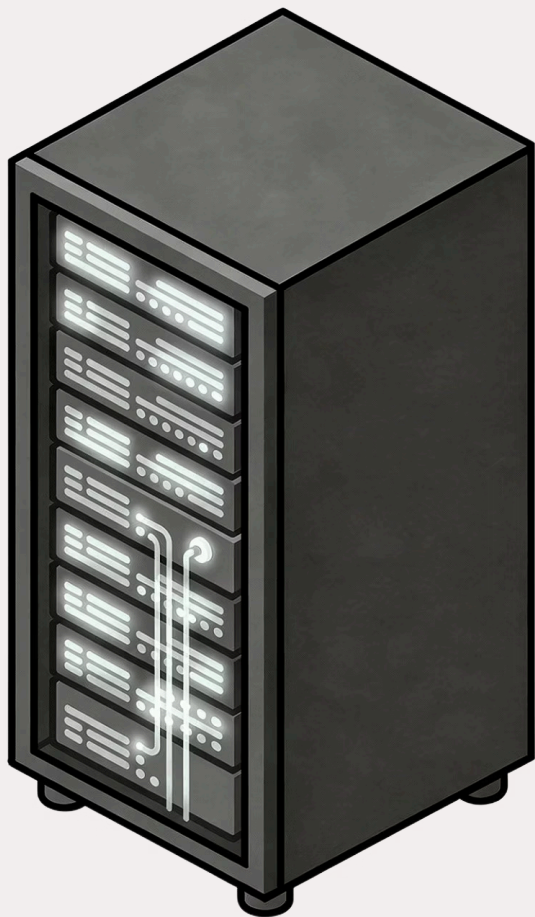




Mastering Airflow: From Monitoring to Observability

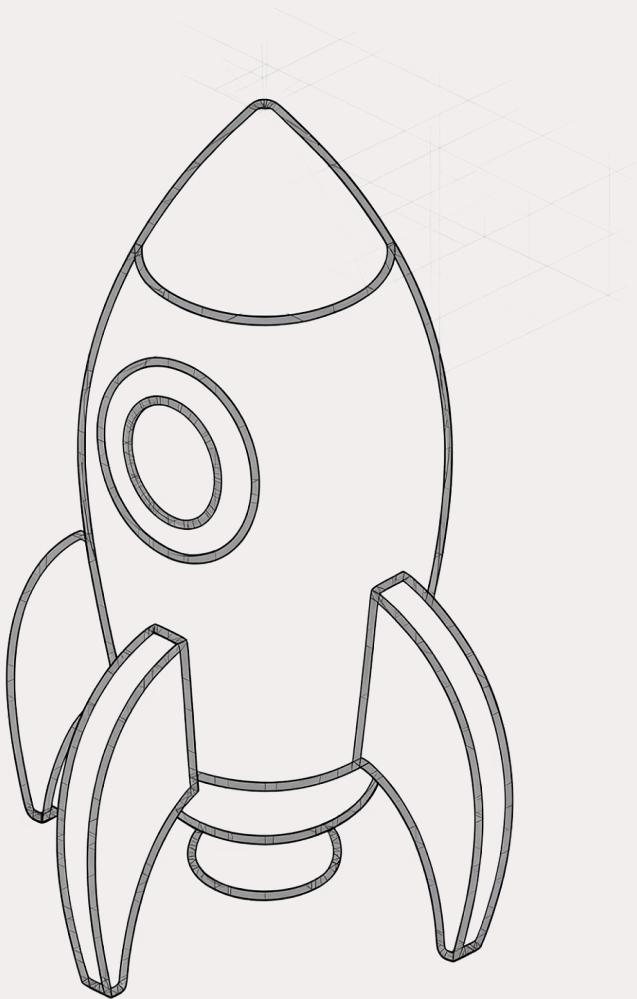
A comprehensive guide to understanding Airflow metrics, SLA management, alerting, and the next generation of orchestration-native observability with Astro Observe.

MODULE 1: AIRFLOW MONITORING

MODULE 2: ASTRO OBSERVE

The Foundation of Airflow Monitoring

Before you can optimize, you must observe. Module 1 lays the groundwork for understanding how Airflow surfaces operational data — from raw metrics and SLA tracking to alerting pipelines and structured log architectures. These building blocks are essential for running reliable, production-grade data workflows.

**1****Metrics Collection****2****SLA Monitoring****3****Alerting Mechanisms****4****Logging Architecture**

Metrics Collection: The Pulse of Your Pipelines

What It Is

Metrics collection means gathering quantitative, time-series data on Airflow's performance and overall health. It answers the question: *is my system behaving as expected?*

Where to Find Them

The Astro UI Analytics page provides real-time insights into DAG and task runs, worker performance, and resource utilization — all in one place.

Key Metrics to Track

**Task Success /
Failure Rates**

**Run Durations (P90,
Average)**

DAG Run Counts

Worker Utilization

Scheduler Health

SLA Monitoring: Ensuring Timely Data Delivery

Service Level Agreements (SLAs) set the contractual expectations for pipeline timeliness and data freshness. When they're missed, business decisions are delayed, ML models run on stale data, and compliance windows close.



Define SLAs

Set pipeline-level expectations for on-time completion and data freshness directly tied to business requirements.



Track & Enforce

Astro Observe continuously evaluates SLA adherence across all data products, flagging deviations in real time.



Business Alignment

Custom business-level SLAs can be created for specific data product delivery milestones, not just task completion.

Alerting Mechanisms: Your Early Warning System

Alerts transform passive monitoring into active defense. Rather than discovering failures after downstream impact, a well-configured alerting system surfaces problems the moment they emerge — giving teams the time to act.

Proactive Notifications

Alerts are triggered by predefined conditions or anomaly detection — not discovered manually after damage is done.

Multi-Channel Delivery

Astro Observe monitors can alert teams via **email**, **Slack**, or custom webhook integrations for maximum coverage.

Configurable Severity

Alerts can be tiered as **Info**, **Warning**, or **Critical** — ensuring the right level of urgency reaches the right people.

Logging Architecture: The Detective's Notebook

Logs are the most granular source of truth in an Airflow environment. While metrics tell you *what* happened, logs tell you *why*. A structured logging architecture is essential for debugging failures, conducting audits, and tracing the exact sequence of events.

Scheduler Logs

Track DAG parsing, scheduling decisions, and heartbeat health.

Worker Logs

Record task execution environments, resource usage, and runtime errors.

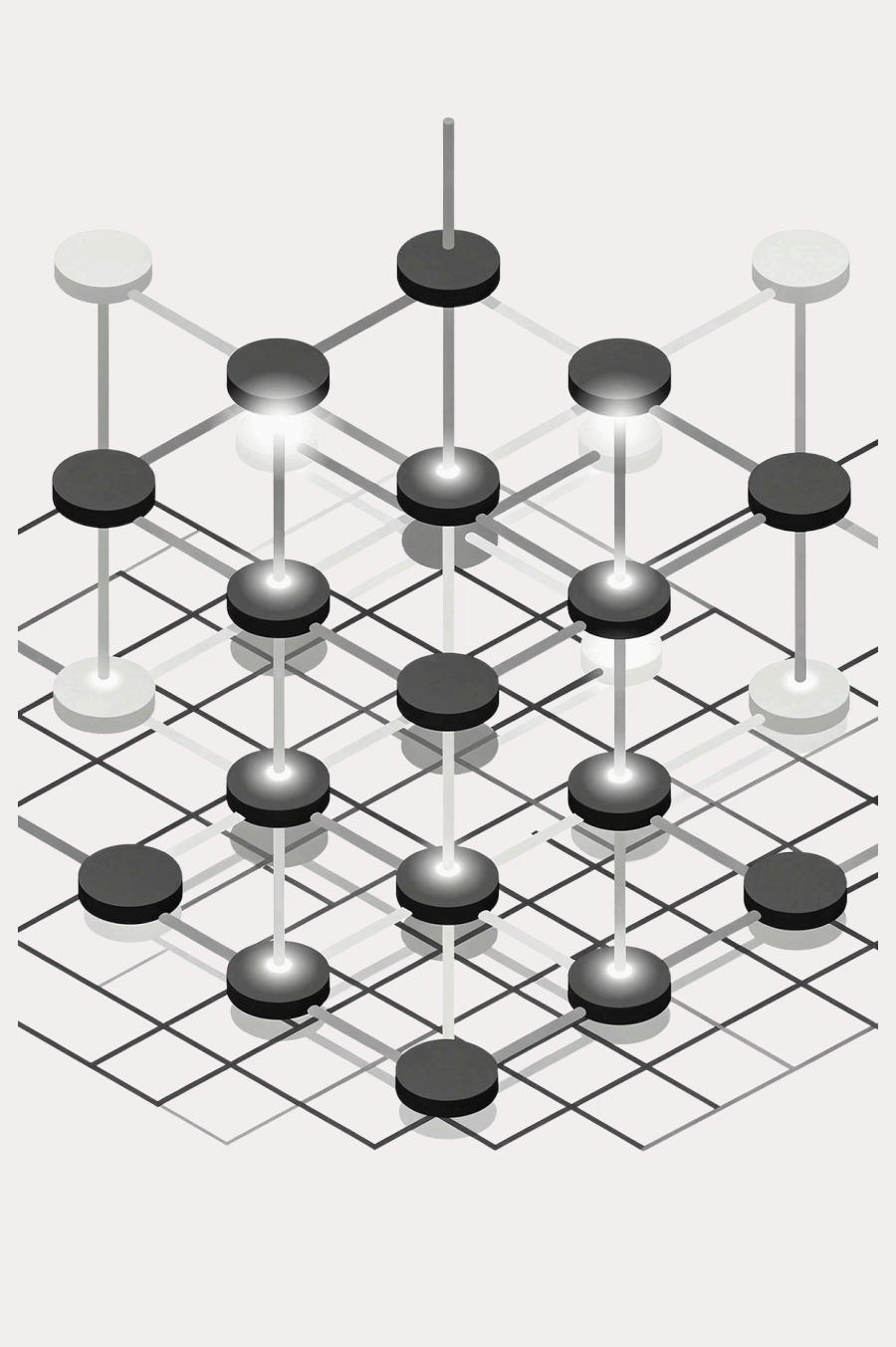
Task Logs

Capture the granular stdout/stderr output of individual task runs.

Webserver Logs

Log API calls, UI interactions, and authentication events.





Visibility Is Key

From raw metrics to structured logs, comprehensive monitoring transforms opaque pipelines into observable, trustworthy systems. Every signal is a data point in the story of your data infrastructure.

Elevating to Orchestration-Native Observability with Astro Observe

Module 2 introduces Astro Observe — a purpose-built observability layer designed to work hand-in-hand with Airflow. It moves beyond infrastructure metrics and into business-level data product health, lineage, and proactive monitoring.

1

Orchestration-Native Observability

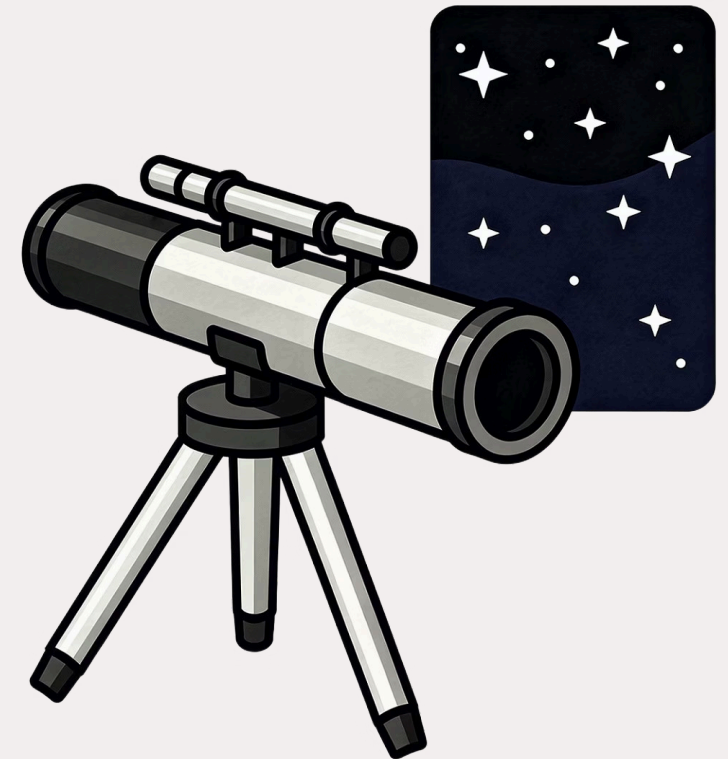
2

DAG Performance Analysis

3

Lineage Visibility

4

Data-Aware Monitoring

Astro Observe: Purpose-Built for Airflow

Orchestration-Native Observability

Unlike generic monitoring tools, Astro Observe is designed from the ground up for Airflow pipelines. It understands DAGs, tasks, and data assets natively — eliminating the integration gap that plagues bolt-on solutions.

A Unified Platform

Astro Observe combines orchestration and observability in a single pane of glass, providing deep visibility into the health, quality, and performance of data products across the entire modern data stack.

Core Capabilities

- Business-level SLA creation and tracking
- Real-time data product health monitoring
- Cross-deployment asset grouping
- Automated lineage graph generation
- Integrated alerting across channels

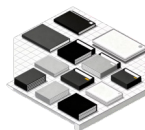
Data Products: Defining Your Business-Critical Assets

A data product is a reusable, curated asset that packages data for a specific business purpose — such as a customer-facing dashboard, a revenue forecasting ML model, or an executive KPI report. Astro Observe elevates these from pipeline outputs to first-class, observable entities.



Automatic Dependency Inference

Astro Observe automatically detects and maps upstream DAG dependencies for each data product, even across multiple deployments.



Asset Grouping

Multiple DAGs and datasets can be grouped into a single data product, providing a business-relevant unit of health tracking.



Real-World Example

A suite of Airflow DAGs populating an executive dashboard becomes one observable data product with a unified SLA and alerting policy.

DAG Performance Analysis: Beyond Basic Metrics

Standard Airflow monitoring tells you if a task failed. Astro Observe tells you *how often*, *how long*, and *what business impact resulted*. This deeper layer of analysis is critical for proactive pipeline optimization.

1

Pipeline-Level Run Metrics

Track failed runs, retry rates, and execution duration trends across entire DAG pipelines — not just individual tasks.

2

Reliability Metrics

Monitor SLA evaluations, alert frequency, and resolution times to measure your pipeline's operational trustworthiness.

3

Silent Failure Detection

Identify jobs that complete without errors but produce incorrect or incomplete data — the most dangerous class of pipeline failure.

Lineage Visibility: Tracing the Data Journey

What Is Data Lineage?

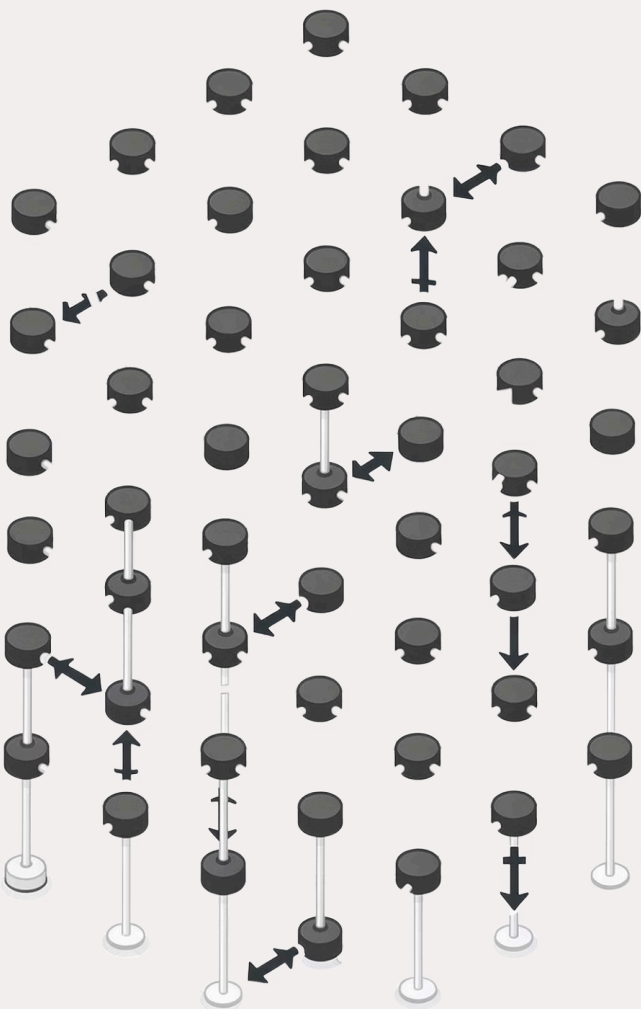
Lineage is a visual, traceable map of how data moves from source systems through transformations to final consumption — answering *"where did this number come from?"*

OpenLineage Integration

Astro Observe leverages the **OpenLineage** standard for comprehensive metadata collection, ensuring compatibility across tools in the modern data stack.

Astro Observe Lineage Features

- Real-time lineage graphs updated on each DAG run
- Relationship mapping between all assets in a data product
- Upstream dependency visualization across deployments
- Direct navigation from a failing asset to its root cause
- Impact analysis: understand what breaks downstream



Unraveling Complexity

Lineage graphs transform abstract data pipelines into a navigable map of cause and effect — making root cause analysis faster and downstream impact analysis immediate.

Data-Aware Monitoring: Understanding the "Why"

Traditional monitoring answers: *"Did the task run?"* Data-aware monitoring answers: *"Did the data arrive correctly, on time, and with the right quality to support business decisions?"* This is the critical distinction Astro Observe makes.



Quality Beyond Status

Monitors evaluate data quality and business impact, not just task execution state.



Proactive Failure Monitor

Automatically alerts when any upstream or final DAG in a data product fails, before downstream consumers are affected.



Lineage-Linked Alerts

Every alert links directly to the lineage view, enabling instant identification of impacted assets and root cause navigation.

Monitors in Astro Observe: Proactive Problem Solving

Monitors are the operational heart of Astro Observe. They run continuously, evaluate pipeline and data product health against defined policies, and take action the moment something deviates from expected behavior.

How Monitors Work

- Continuously assess data product and pipeline health against defined criteria
- Evaluate SLA adherence on every pipeline run
- Detect failures in upstream dependencies before they surface downstream

Alert Severity Levels

Info

Informational notices, no immediate action required.

Warning

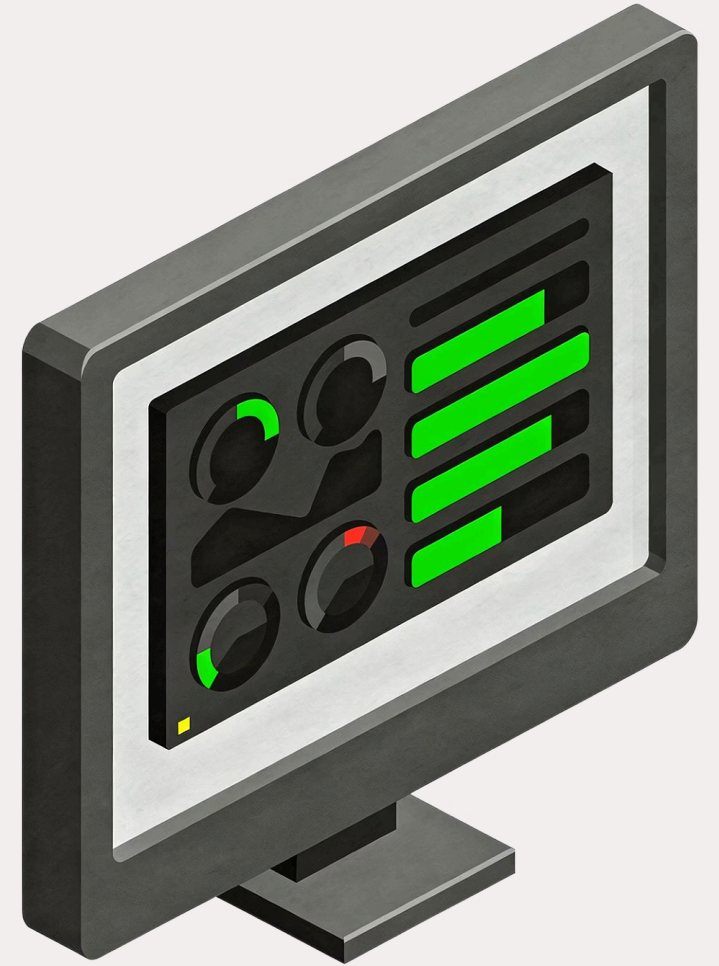
Potential degradation; investigate soon.

Critical

Immediate action required; business impact likely.

Proactive Insights, Faster Resolutions

When every monitor, alert, and lineage view is connected in a single platform, data teams spend less time hunting for problems and more time solving them before anyone notices.



KEY BENEFITS

Key Benefits of Astro Observe



Reduced MTTR

Faster diagnosis through lineage-linked alerts and contextual error details dramatically cuts Mean Time To Resolution.



Improved Reliability

Continuous SLA evaluation ensures data products consistently meet the business expectations set by stakeholders and compliance requirements.



Enhanced Efficiency

Performance analysis pinpoints bottlenecks, enabling targeted optimization of resource allocation and pipeline execution.



Business Alignment

Directly connects pipeline health to business outcomes, making data infrastructure health legible and actionable for stakeholders.

Real-World Impact: Preventing Data Disasters

Consider a critical data product — an executive revenue dashboard — that misses its morning SLA due to a failure three hops upstream in the pipeline.

Without Astro Observe

- Delayed reports reach executives with stale or missing data
- ML models run predictions on yesterday's incomplete dataset
- Compliance windows missed; manual incident investigation begins
- Hours pass before the root cause DAG is even identified

With Astro Observe

- Immediate Critical alert fires the moment the upstream DAG fails
- Alert links directly to the failing task and lineage graph
- All downstream impacted products are instantly visible
- Rapid remediation — often before any consumer notices an issue



Reactive vs. Proactive

The difference between discovering a failure at 9 AM in a stakeholder's inbox versus resolving it silently at 3 AM is the difference between reactive monitoring and orchestration-native observability.

The Future of Airflow Operations: Orchestration Meets Observability

The evolution from ad-hoc monitoring to strategic data operations is not a tool upgrade — it's a fundamental shift in how data teams operate and deliver value.



Reactive Troubleshooting

Manual log hunting, post-failure investigation, and stakeholder-reported issues.



Active Monitoring

Metrics, alerts, and SLAs provide timely notifications — but still require manual diagnosis.



Orchestration-Native Observability

Astro Observe delivers full-context, lineage-linked, business-aligned data operations — proactive, automated, and continuous.



Embrace data observability with Astro Observe to build a competitive advantage on reliable, transparent, and business-aligned data pipelines.